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$$f(x) = \frac{1}{1 + e^{2x}}$$

$$f'(x) = -\frac{2e^{2x}}{(1 + e^{2x})^2}$$

$$\begin{aligned} f''(x) &= -\frac{4e^{2x}}{(1 + e^{2x})^2} + \frac{8e^{4x}}{(1 + e^{2x})^3} \\ &= \frac{4(e^{2x} - 1)e^{2x}}{(e^{2x} + 1)^3} \end{aligned}$$

References